

Review

A Review of Traditional and Advanced MPPT Approaches for PV Systems Under Uniformly Insolation and Partially Shaded Conditions

Mustafa Sacid Endiz ¹, Göksel Gökkuş ² , Atıl Emre Coşgun ^{3,*}  and Hasan Demir ³ ¹ Faculty of Engineering, Necmettin Erbakan University, Konya 42090, Turkey; msendiz@erbakan.edu.tr² Faculty of Engineering and Architecture, Nevşehir Hacı Bektaş Veli University, Nevşehir 50300, Turkey; gokselgokkus@nevsehir.edu.tr³ Faculty of Engineering, Aksaray University, Aksaray 68100, Turkey; hasandemir@aksaray.edu.tr

* Correspondence: atilemrecosgun@aksaray.edu.tr

Abstract: Solar photovoltaic (PV) is a crucial renewable energy source that converts sunlight into electricity using silicon-based semiconductor materials. However, due to the non-linear characteristic behavior of the PV module, the module's output power varies according to the solar radiation and the ambient temperature. To address this challenge, maximum power point tracking (MPPT) techniques are employed to extract the maximum amount of power from the PV modules. This paper offers a comprehensive review of widely used traditional and advanced MPPT approaches in PV systems, along with current developments and future directions in the field. Under uniform insolation, these methods are compared based on their strengths and weaknesses, including sensed parameters, circuitry, tracking speed, implementation complexity, true MPPT, accuracy, and cost. Additionally, MPPT algorithms are evaluated in terms of their performance in reaching maximum power point (MPP) under partial shading condition (PSC). Existing research clearly demonstrates that the advanced techniques exhibit superior efficiency in comparison to traditional methods, although at the cost of increased design complexity and higher expenses. By presenting a detailed review and providing comparison tables of widely used MPPT techniques, this study aims to provide valuable insights for researchers and practitioners in selecting appropriate MPPT approaches for PV applications.

Keywords: renewable energy; solar power; PV array; partial shading; MPPT techniques



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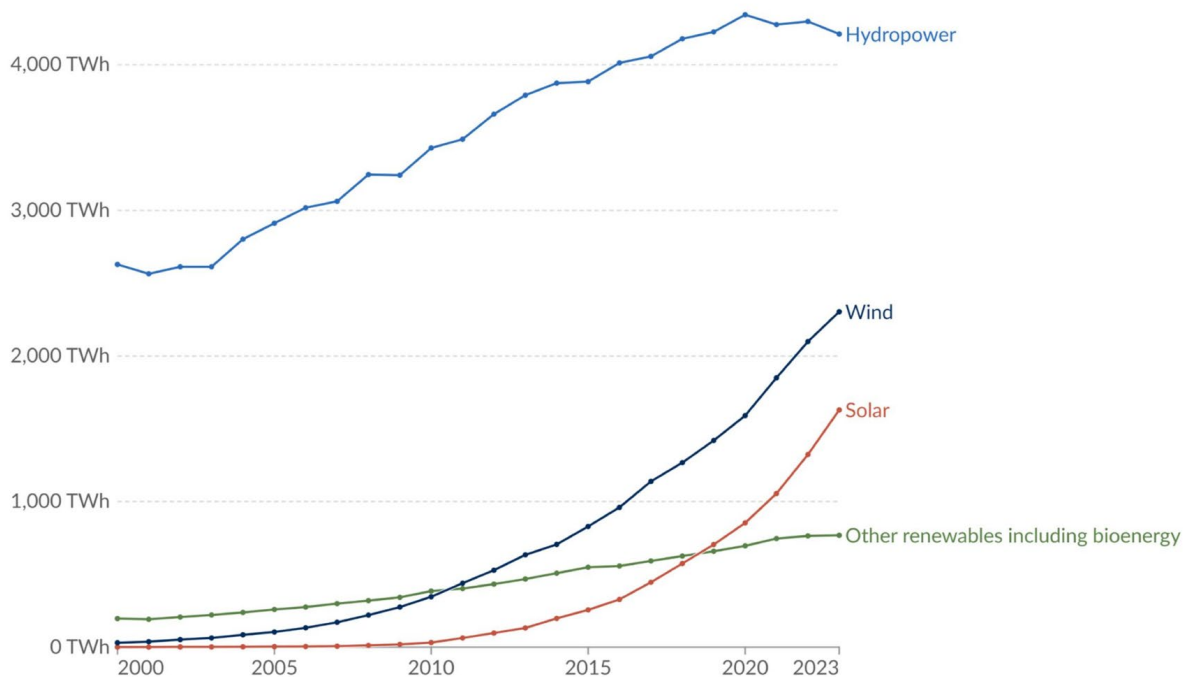
1. Introduction

Renewable energy technologies are increasingly becoming viable options for achieving global electricity access efficiently and consistently. Renewable energy systems have become essential alternatives to conventional energy sources due to increasing environmental concerns and the limited availability of fossil fuel supplies, playing a crucial role in advancing sustainable development [1–7]. Solar energy is one of the most abundant and widely available renewable energy sources. Photovoltaic (PV) systems and solar thermal technologies may both harvest the sun's energy [8–14]. The widespread availability and scalability of solar energy make it particularly appealing, as it can be used in a variety of applications, from modest residential installations to large-scale solar farms. Solar power has emerged as a prominent participant in the global transition to renewable energy due to the advancement of technology and falling costs [15–19]. The global renewable energy generation by source in 2023 is depicted in Figure 1 [20]. The share of solar energy in

production, among renewable energy sources, began to develop significantly in the early 2000s and has continued to grow rapidly with a sharp increase since 2010, approaching the 2000 TWh level.

Modern renewable energy generation by source, World

Measured in terawatt-hours¹.



Data source: Ember (2024); Energy Institute - Statistical Review of World Energy (2024)
OurWorldInData.org/renewable-energy | CC BY

1. Watt-hour: A watt-hour is the energy delivered by one watt of power for one hour. Since one watt is equivalent to one joule per second, a watt-hour is equivalent to 3600 joules of energy. Metric prefixes are used for multiples of the unit, usually: - kilowatt-hours (kWh), or a thousand watt-hours. - Megawatt-hours (MWh), or a million watt-hours. - Gigawatt-hours (GWh), or a billion watt-hours. - Terawatt-hours (TWh), or a trillion watt-hours.

Figure 1. The global distribution of power generation by renewable energy source in 2023.

Solar energy is converted into electricity using PV cells. In a typical PV system, multiple cells are electrically connected in series and parallel configurations to form PV strings, allowing for scalable design and power requirements [15,21]. However, the ability of solar energy systems to capture and convert solar irradiance into usable electrical power is a critical factor in their efficiency. Solar radiation and temperature are key factors in determining the efficiency of PV modules. High radiation levels increase energy production, while high temperatures can reduce efficiency due to increased electrical resistance. The optimal power generation occurs when sunlight is abundant and temperature remains within an ideal range to minimize efficiency losses [10,22–24]. In order to enhance the efficiency of PV systems, the use of Maximum Power Point Tracking (MPPT) is implemented [25,26]. MPPT is a method employed to optimize the energy extracted from solar modules by ensuring they operate at the maximum power point (MPP). The MPP is the specific voltage and current combination at which a PV module delivers the highest power output as shown in Figure 2. The MPP is not constant and changes with variations in solar irradiance and temperature, reflecting the dynamic nature of these environmental factors. MPPT algorithms are developed to consistently modify the electrical operating point of the PV modules to align with the MPP, thereby improving overall energy capture and efficiency [27–32].

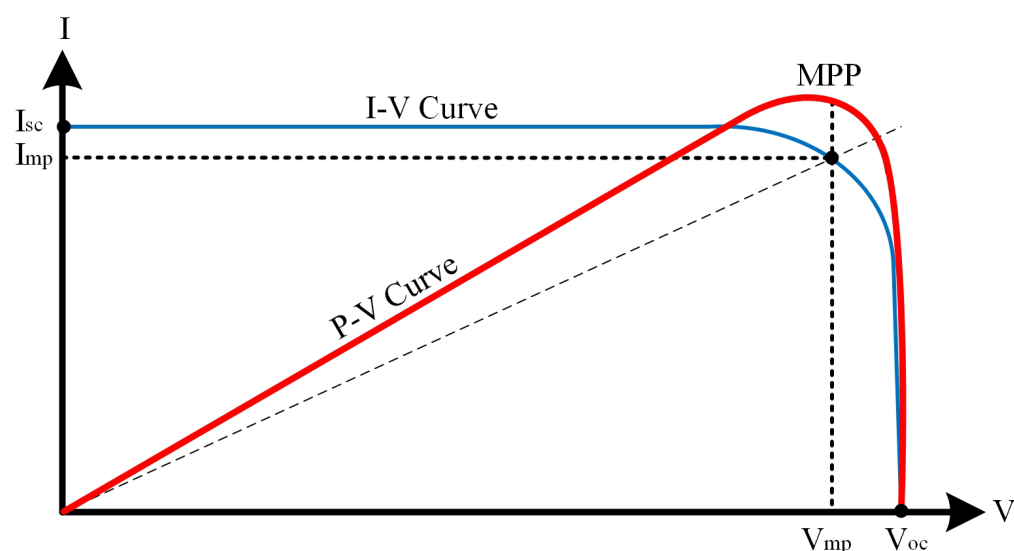


Figure 2. Current–Voltage characteristic of a PV cell.

Figure 3 depicts the operating block diagram of the MPPT in a PV system. It includes the interactions between the PV modules, the DC-DC converter, the MPPT controller, and the load. The figure outlines how the MPPT controller continuously monitors the PV system's generated voltage and current values, processes these data to identify the MPP, and sends appropriate control signals to the DC-DC converter to adjust the duty cycle, ensuring the system operates at MPP.

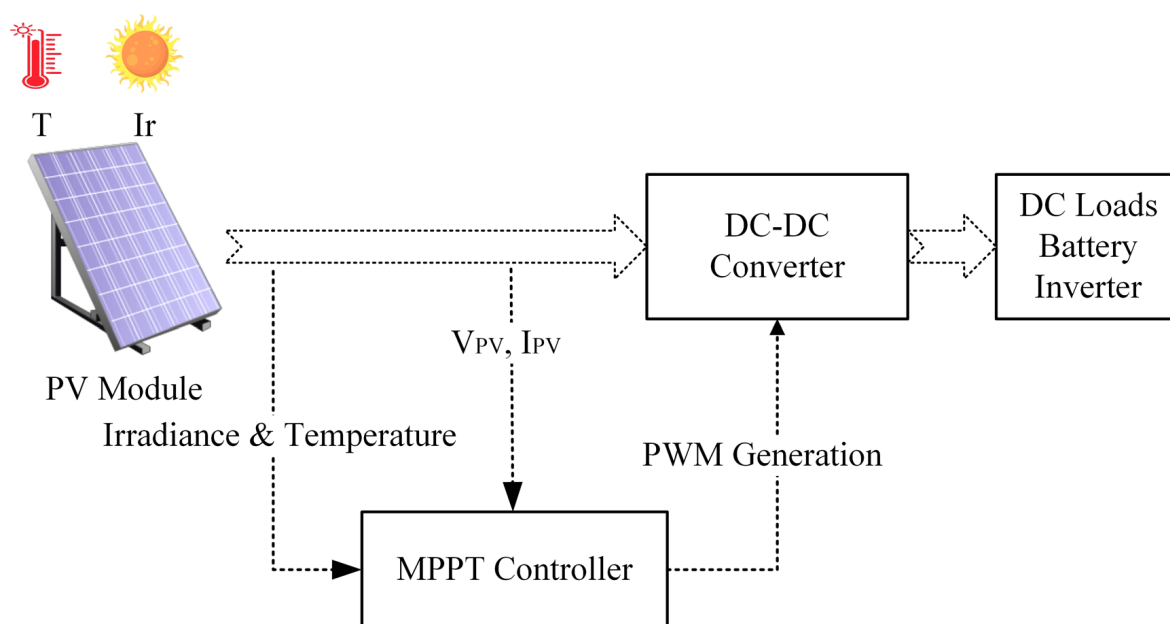


Figure 3. Block diagram of a PV system with MPPT.

As shown in Table 1, MPPT algorithms can be mainly categorized into two groups based on their tracking characteristics: traditional and advanced approaches.

Table 1. Classifications of MPPT algorithms.

Category		Sub-Category	Acronym		
Advanced MPPT algorithms	Traditional MPPT algorithms	Perturb and Observe	P&O		
		Incremental Conductance	IC		
		Constant Voltage	CV		
		Lookup Table	LT		
		Hill Climbing	HC		
	Smart techniques	Fuzzy Logic Controller	FLC		
		Artificial Neural Network	ANN		
		Fibonacci Series	FS		
	Metaheuristic techniques	Optimization-based techniques	Particle Swarm Optimization	PSO	
			Grey Wolf Optimization	GWO	
			Ant Colony Optimization	ACO	
			Artificial Bee Colony	ABC	
			Cuckoo Search Optimization	CSO	
		Bio-inspired techniques	Firefly Optimization	FO	
			Flower Pollination	FP	
			Genetic Algorithm	GA	
			Hybrid techniques	Particle Swarm Optimization and Perturb and Observe	
		Grey Wolf Optimization and Perturb and Observe		GWO-P&O	
		Fuzzy Logic Controller and Perturb and Observe		FLC-P&O	
		Genetic Algorithm and Perturb and Observe		GA-P&O	
		Adaptive Neuro-Fuzzy Inference System		ANFIS	

The development of MPPT algorithms demonstrates a significant progression from simple techniques to advanced approaches. Early traditional MPPT techniques are founded on clear principles and have been extensively used because of their simplicity and low cost in stable conditions. However, their performance may be affected in dynamic or unpredictable scenarios, resulting in less-than-optimal achievements. With advancements in computational power and the growing understanding of complex systems, there has been a shift in focus toward advanced MPPT algorithms. The methods described use machine learning and optimization techniques to improve the accuracy and adaptability of tracking. The latest advancements in MPPT algorithms are focusing on hybrid approaches that combine traditional methods with metaheuristic MPPT techniques. In a review study, a concise classification and evaluation of various MPPT methods for PV systems, highlighting their efficiency, performance, complexity, and tracking speed are presented. It categorizes the methods into conventional, advanced, and hybrid approaches [25]. Another study provides a comprehensive review of MPPT algorithms for PV systems, categorizing them into measurement-based, calculation-based, intelligent schemes, and hybrid methods. It compares the performance, applications, advantages, and disadvantages of various algorithms, offering valuable insights for selecting the most suitable one [33]. In another study, the methods used for MPPT in PV systems are reviewed. These methods are classified into

conventional, intelligent, optimization, and hybrid techniques. A comparison is made of the different methods based on criteria such as tracking speed, efficiency, cost, stability, and implementation complexity. The review highlights that hybrid techniques, while more efficient than conventional methods, are more complex in design and costly [34]. In a study, a survey of MPPT methods for PV systems is presented, focusing on their ability to continually maximize PV array output power, which depends on solar radiation and cell temperature. MPPT methods are classified into two categories: conventional methods and advanced methods. The paper analyzes, simulates, and evaluates the performance of a PV power supply system under varying meteorological conditions [35]. Another study provides an organized and in-depth review of MPPT techniques for PV systems, categorizing them into classical, intelligent, optimization, and hybrid methods. The review considers various selection benchmarks not typically discussed in existing literature, summarizing the advantages and disadvantages of each technique [36].

This paper provides an in-depth review of the most used traditional and advanced MPPT techniques for PV systems under uniform insolation and partial shading condition (PSC). It evaluates these methods by comparing their strengths and weaknesses in terms of sensed parameters, circuitry, tracking speed, implementation complexity, true MPPT, accuracy, and cost. The rest of the paper is structured as follows: Sections 2 and 3 explain the effect of shading on PV systems and the selection parameters of the MPPT strategies. Sections 4 and 5 focus on the traditional and advanced MPPT techniques in PV systems. Section 6 discusses the findings of various MPPT algorithms comprehensively, offering comparison tables highlighting the efficiency of each algorithm. Section 7 outlines current trends and directions for future research, while Section 8 offers conclusions.

2. The Effect of Shading on PV Systems

Shading is one of the most significant factors that can impact the performance of PV systems [37,38]. The power output of a PV module can be represented by a power–voltage (P–V) curve. When the PV array is partially shaded, the curve becomes non-linear, with multiple peaks that represent different local maximum power points (Local MPP) instead of a single global maximum power point (Global MPP), as shown in Figure 4. The shaded part of the module acts like a resistor, absorbing energy from the unshaded cells, effectively lowering the overall power produced by the PV array. This effect is more significant in series-connected PV systems, where the generated current is limited by the PV module with the lowest current.

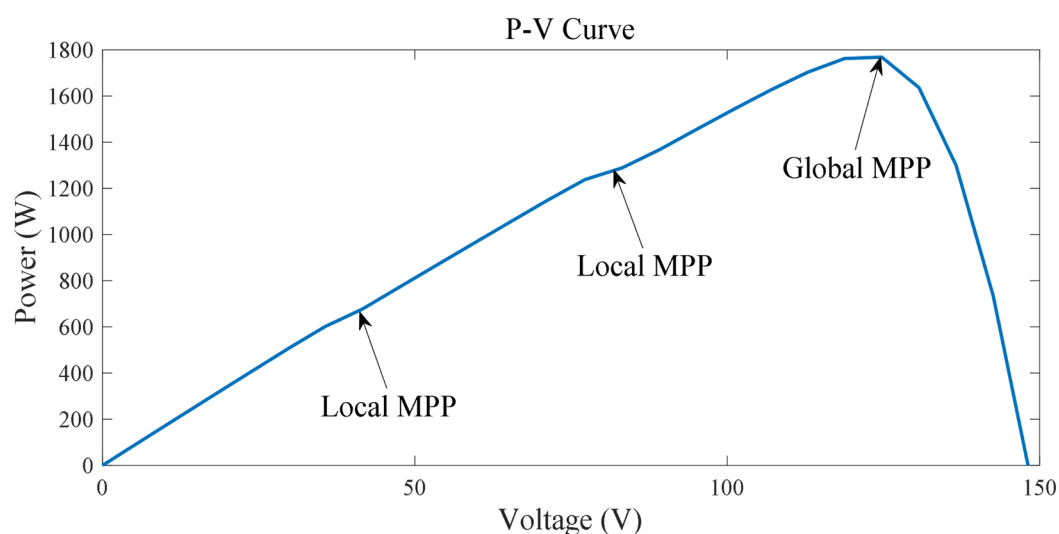


Figure 4. Global and local maximum power points of a PV array under PSC.

3. The Selection Parameters of the MPPT Algorithms

The evaluation parameters are selected to compare MPPT algorithms since these factors determine the performance, efficiency, and cost-effectiveness of the algorithm. Each of these parameters covers a different aspect of how the MPPT functions and how it integrates into the overall design of a PV system. The selected parameters are intended to comprehensively evaluate both traditional and advanced algorithms [39]. The most widely used parameters for evaluating MPPT algorithms are as follows: sensed parameters, circuitry, tracking speed, implementation complexity, true MPPT, accuracy, and cost.

3.1. Sensed Parameters

Sensed parameters are the physical quantities that need to be measured in order to implement the MPPT algorithm. Common parameters include the output voltage and current of the PV module, solar radiation, and temperature used to track the maximum operating point. The type of parameters sensed affects the algorithm's complexity and response time.

3.2. Circuitry

Circuitry refers to the hardware required to implement the MPPT algorithm. The complexity and efficiency of the circuitry can influence the overall performance and cost of the MPPT system. MPPT algorithms can be implemented using analog or digital circuitry. Analog implementations are simpler and faster but may lack precision and scalability. Digital implementations, often using microcontrollers or DSPs, offer flexibility, higher precision, and easier integration with other systems.

3.3. Tracking Speed

Tracking speed is the rate at which the MPPT algorithm adjusts to changes in response to changing environmental conditions such as irradiance and temperature to maintain optimal power output. Faster tracking minimizes the loss of power during transient changes. In PV systems, environmental factors like shading or temperature fluctuations can cause rapid changes in the amount of power generated by the PV modules. The MPPT algorithm must be able to quickly adjust the operating point (voltage and current) to match the new optimal power point.

3.4. Implementation Complexity

Implementation complexity refers to the ease or difficulty of implementing the MPPT algorithm in hardware or software. Simpler algorithms are easier to implement but may be less efficient under certain conditions. More complex algorithms require more advanced coding, hardware, and tuning. The tradeoff involves balancing performance with the development effort, debugging, and system integration.

3.5. True MPPT

A true MPPT algorithm is one that always tracks the actual MPP of the PV module under all conditions, regardless of external conditions. Some algorithms approximate the maximum power but may not always reach it due to their limitations, such as slow tracking or being affected by local maxima. True MPPT techniques adapt more robustly to rapidly changing irradiance or temperature and can avoid power loss due to inefficient tracking.

3.6. Accuracy

Accuracy in MPPT refers to how precisely the MPPT algorithm tracks the true MPP. High accuracy means the system operates close to the actual MPP, maximizing energy harvest. Some algorithms achieve high accuracy at the cost of increased complexity or slower tracking, while others may be less accurate but simpler or faster.

3.7. Cost

The cost is the financial expense involved in implementing the MPPT algorithm, which includes both hardware and software requirements. More complex algorithms may require more expensive components or higher computational resources. Digital implementations may require specialized processors, sensors, and converters, driving up costs. Analog circuits are cheaper but may be less flexible or efficient. The cost also includes power losses due to less efficient tracking.

4. Traditional MPPT Techniques

Traditional MPPT techniques are used to optimize power output in PV systems. These techniques are characterized by their simple implementation and relatively low computational requirements.

4.1. Perturb and Observe Algorithm

The Perturb and Observe (P&O) MPPT algorithm is a widely used technique for optimizing the power output of PV systems. It operates by periodically perturbing the operating voltage of the PV system and observing the change in power output [40]. If the power increases, the algorithm continues perturbing in the same direction; if the power decreases, it reverses the perturbation direction. The main advantage of P&O is its simplicity and ease of implementation, making it suitable for a wide range of applications. However, its disadvantages include potential inefficiencies in the presence of partial shading and its tendency to oscillate around the MPP [41,42]. The flowchart of P&O is shown in Figure 5, where $V(t)$ is the module voltage, $I(t)$ is the module current, $P(t)$ is the module output power, $D(t)$ is the duty cycle of the DC-DC converter, and ΔD is the variation in the duty cycle.

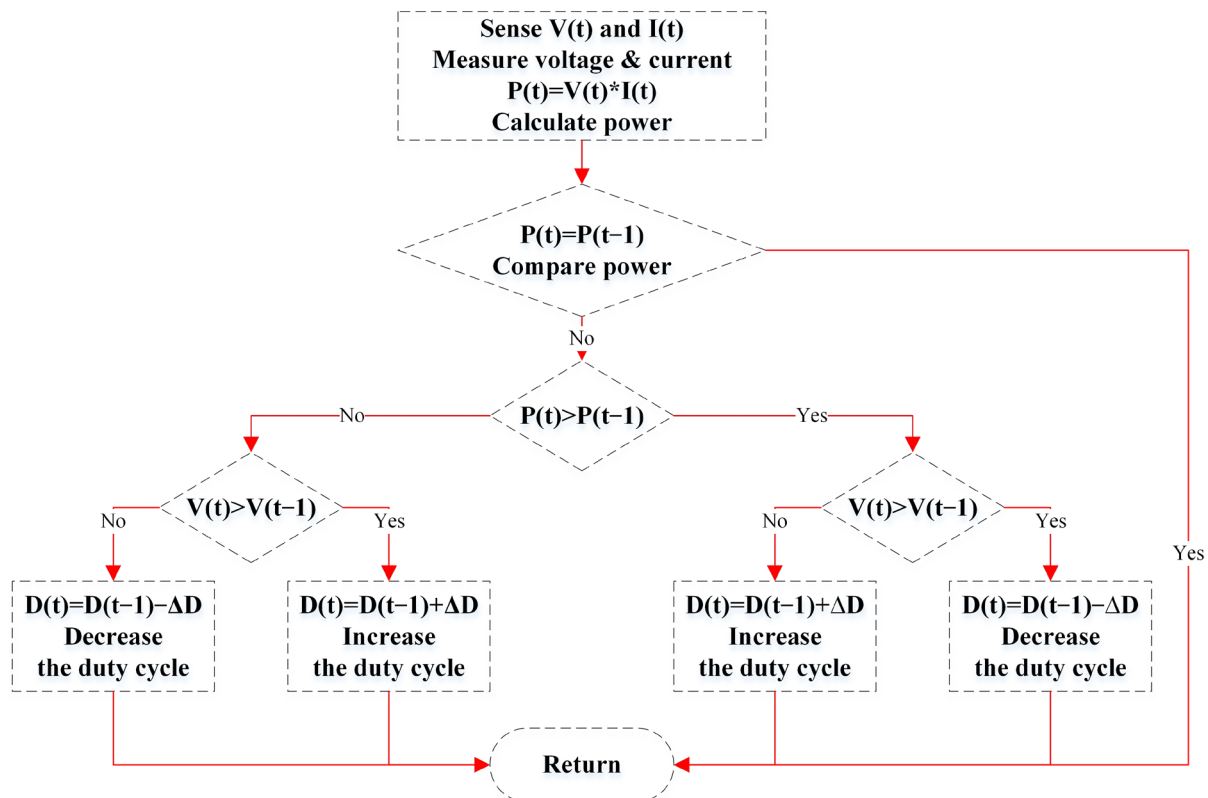


Figure 5. Flowchart of the P&O MPPT.

4.2. Incremental Conductance Algorithm

The Incremental Conductance (IC) MPPT algorithm is an advanced technique designed to optimize the power output of PV systems by analyzing the rate of change in the power output with respect to voltage. The algorithm calculates the incremental conductance (the change in current divided by the change in voltage) and compares it to the instantaneous conductance (current divided by voltage) to determine whether to increase or decrease the voltage to reach the MPP [43–46]. One of the key advantages of the IC algorithm is its ability to accurately track the MPP even under rapidly changing environmental conditions. However, IC may still exhibit some oscillation around the MPP in highly dynamic environments [47]. Figure 6 illustrates the flowchart of the IC MPPT, where dP is the power variation, dV is the voltage variation, dI is the current variation, and V_{ref} is the reference voltage. The following equations are used to explain the IC method.

$$\frac{dP}{dV} = \frac{d(I \times V)}{dV} = I + V \times \frac{dI}{dV} \quad (1)$$

$$\frac{dP}{dV} = 0 \text{ or } \frac{dP}{dV} > 0 \text{ or } \frac{dP}{dV} < 0 \quad (2)$$

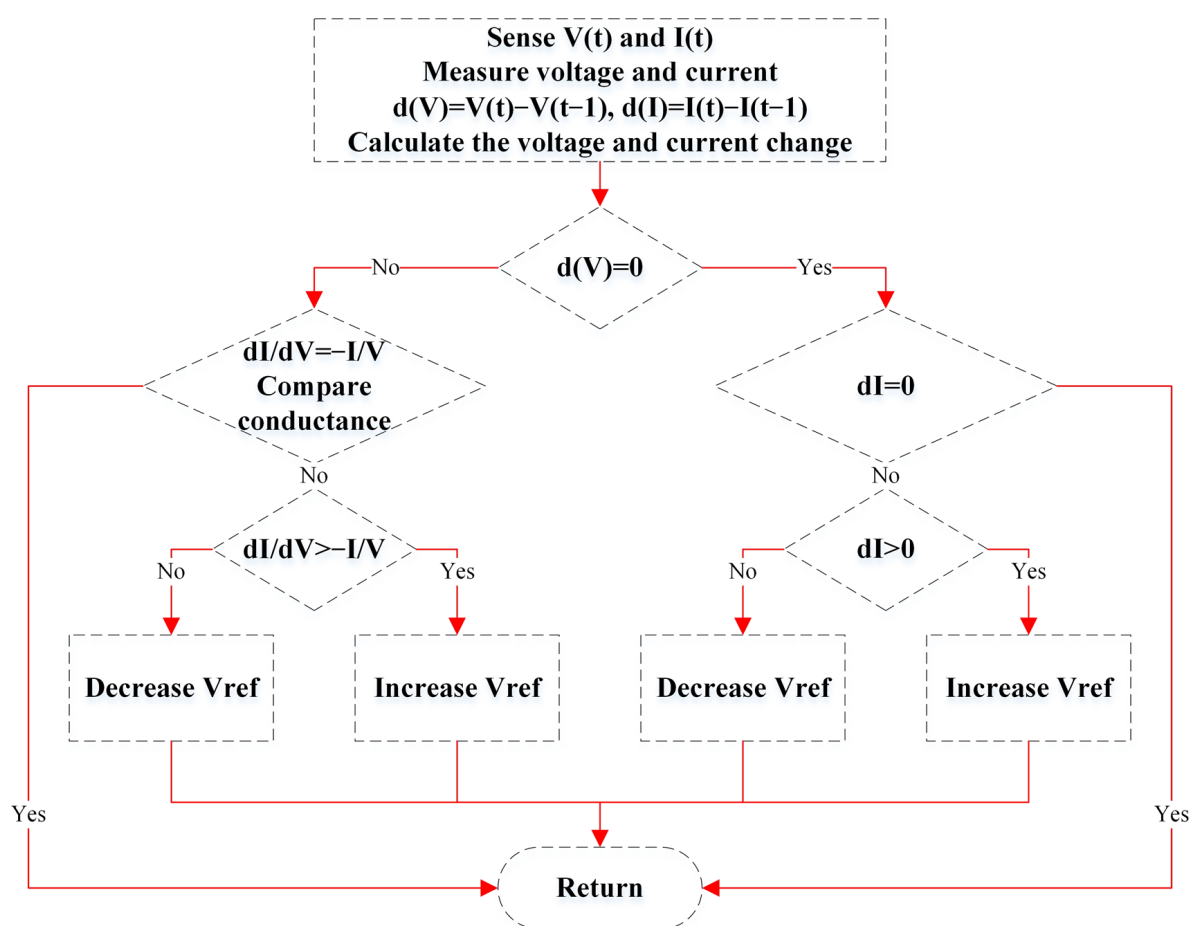


Figure 6. Flowchart of the IC MPPT.

4.3. Constant Voltage Algorithm

The Constant Voltage (CV) MPPT algorithm is the simplest approach used to maximize the power output of PV systems by maintaining the system voltage at a fixed value close to the MPP. This fixed voltage is determined based on the characteristics of the PV modules

and is typically set to a value where the modules are expected to deliver optimal power under standard conditions [48]. Figure 7 illustrates the flowchart of the CV MPPT where the PV module voltage is measured and compared with the voltage of the module at the MPP (V_{MPP}), any error detected is used to adjust the module voltage accordingly using a constant (C). The main advantage of the CV algorithm is its simplicity, which makes it cost-effective and suitable for systems with less demanding performance requirements. However, its primary disadvantage is that it lacks adaptability to varying environmental conditions such as changes in irradiance and temperature, which can lead to suboptimal power extraction compared to more dynamic MPPT algorithms like P&O or IC. Consequently, while CV can be effective in stable environments, it may not perform as well in situations with significant fluctuations in sunlight or temperature [49].

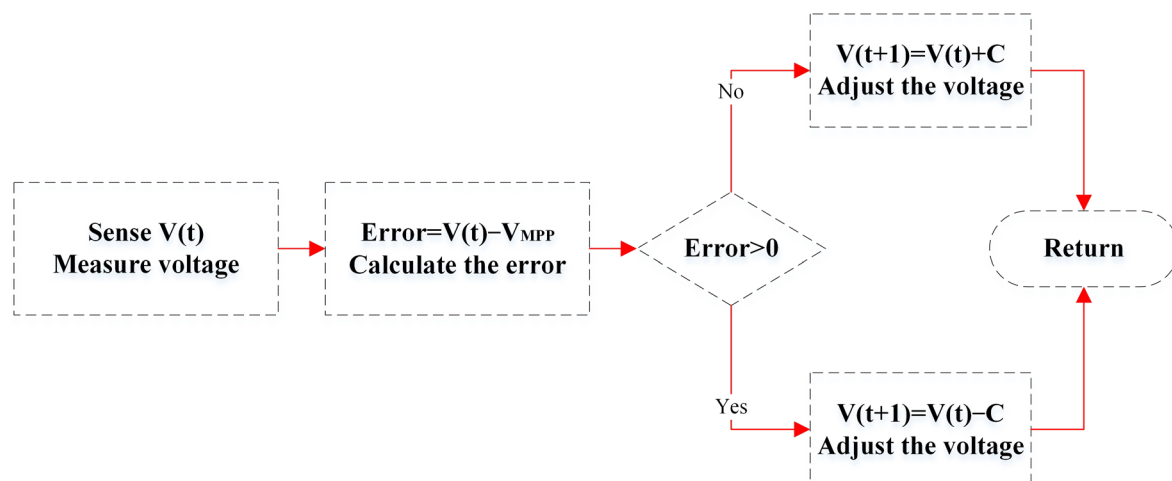


Figure 7. Flowchart of the CV MPPT.

4.4. Lookup Table-Based Algorithm

The Lookup Table (LT) MPPT algorithm relies on precomputed data stored in a lookup table to optimize the power output of PV systems. This approach involves creating a table that maps different environmental conditions, such as irradiance and temperature, to the corresponding optimal voltage or current settings for maximum power extraction. During operations, the system uses this table to determine the most efficient operating point based on real-time measurements [50]. One of the main advantages of the LT algorithm is its fast response time, as it eliminates the need for complex real-time computations by using predefined data. However, its primary disadvantage is that it requires accurate and extensive data collection to build the lookup table, and its performance is limited by the comprehensiveness and accuracy of these data. If the environmental conditions differ greatly from those used to build the table, the algorithm might not adjust properly to conditions that fall outside of its range, thereby resulting in inefficiencies [51]. Figure 8 shows the flowchart of the LT MPPT.

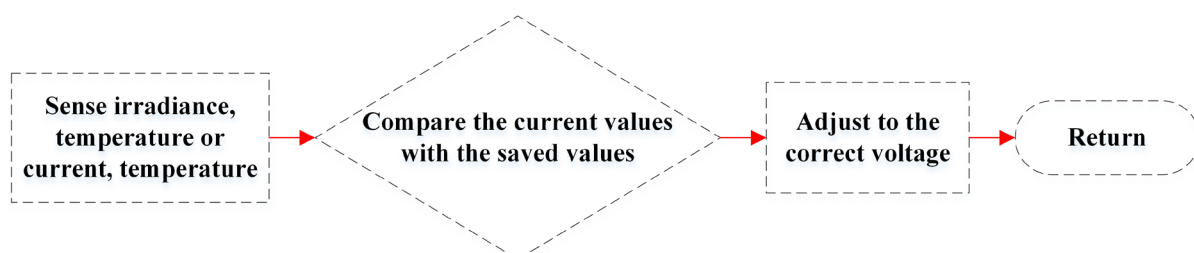


Figure 8. Flowchart of the LT MPPT.

4.5. Hill Climbing Algorithm

Hill Climbing (HC) MPPT is another traditional technique used in solar power systems to optimize energy output by continuously adjusting the operating point to find the maximum power available from a PV module. In this algorithm, the duty cycle is adjusted directly, unlike P&O, which changes the current or voltage. Thus, the power output changes are observed to find the direction that maximizes module power until reaching a peak [52,53]. The main advantages of HC MPPT include its simplicity and effectiveness in variable sunlight conditions. However, it can be less efficient in rapidly changing weather conditions and may suffer from oscillations around the MPP, potentially leading to reduced performance in comparison to more advanced MPPT algorithms [54]. A comparison of the traditional MPPT algorithms is shown in Table 2, where V is voltage, I is current, G is solar radiation, T is temperature sensor, A is analog, and D is digital.

Table 2. Comparison of traditional MPPT algorithms [34,53,55].

Tracking Algorithm	Sensed Parameters	Circuitry	Tracking Speed	Implementation Complexity	True MPPT	Accuracy	Cost
P&O	V, I	A/D	Slow	Low	Yes	Medium	Low
IC	V, I	D	Medium	Medium	Yes	Medium	Medium
CV	V	A	Slow	Low	No	Low	Low
LT	G, T or I, T	D	Medium	Low	Varies	High	Low
HC	V, I	D	Slow	Low	Yes	Medium	Low

5. Advanced MPPT Techniques

Advanced MPPT approaches are mainly characterized by their ability to learn and adapt to complex and dynamic environmental conditions, providing improved accuracy and efficiency in tracking the MPP compared to traditional techniques. They can be categorized into three main types: smart techniques, metaheuristic techniques, and hybrid techniques. Each category offers unique advantages and can be selected based on the specific requirements.

5.1. Smart MPPT Techniques

Smart MPPT techniques offer significant advantages in terms of accuracy, adaptability, and efficiency, making them increasingly important for optimizing PV systems in increasingly complex and variable conditions.

5.1.1. Fuzzy Logic Controller Algorithm

The Fuzzy Logic Controller (FLC) MPPT algorithm uses a rule-based system to optimize the power output of PV systems by handling imprecise and uncertain data. It operates on the principles of fuzzy logic, where the control actions are determined by a set of “if-then” rules that model the relationships between input variables, such as voltage and current, and the desired output [56]. As shown in Figure 9, the FL technique has three stages. The first stage, fuzzification, converts numerical data into linguistic values using membership functions. In the second stage, the rule table processes these entries to make a decision. The final stage, defuzzification, converts linguistic data back into clear numerical values. The FL algorithm evaluates the difference between the current power and the power from the previous time step, as well as the rate of change in power, using fuzzy rules. The key advantage of the FLC algorithm is its flexibility and robustness in managing varying conditions and uncertainties, such as fluctuating irradiance or temperature changes, without needing a precise mathematical model of the system. This adaptability allows it to effectively handle complex scenarios and partial shading. The primary drawback is the complexity of the design and tuning of the fuzzy rules, which may necessitate a high level of expertise [57,58]. The functions that indicate error (E) and change in error (ΔE), with their corresponding equations shown below, are commonly

used as inputs to the FL algorithm. In the given equations, P and V are module output power and module voltage, (n) represents the current value, while $(n - 1)$ indicates the value from the previous sampling period. Based on these equations, the control signal is generated to maximize power output.

$$E(n) = \frac{P(n) - P(n-1)}{V(n) - V(n-1)} \quad (3)$$

$$\Delta E = E(n) - E(n-1) \quad (4)$$

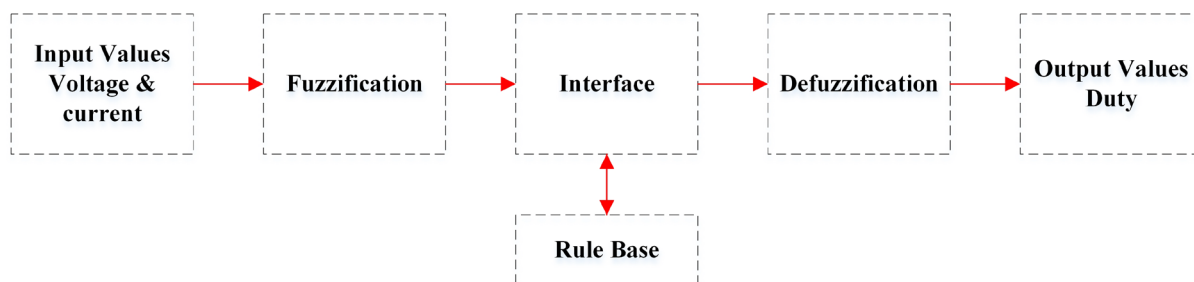


Figure 9. Flowchart of the FLC MPPT.

5.1.2. Artificial Neural Network Algorithm

The Artificial Neural Network (ANN) MPPT algorithm uses machine learning models to predict and track the MPP of PV systems. ANNs are designed to simulate the way the human brain processes information, learning complex patterns from historical and real-time data to make informed decisions about the optimal operating conditions [59,60]. A typical ANN architecture is depicted in Figure 10. The input layer processes raw data through hidden layers, which extract and transform features, capture complex patterns, and generate predictions based on this processed information. Input data for neural networks can include characteristics of PV modules, while the output data regulate the DC-DC converter operation. One of the main advantages of the ANN MPPT algorithm is its ability to handle non-linear and dynamic environments with high accuracy, as it can learn from a wide range of data and adapt to changing conditions such as varying irradiance and temperature. However, the effectiveness of ANNs depends heavily on the quality and quantity of the training data, and they may require extensive retraining if system conditions change substantially [61,62].

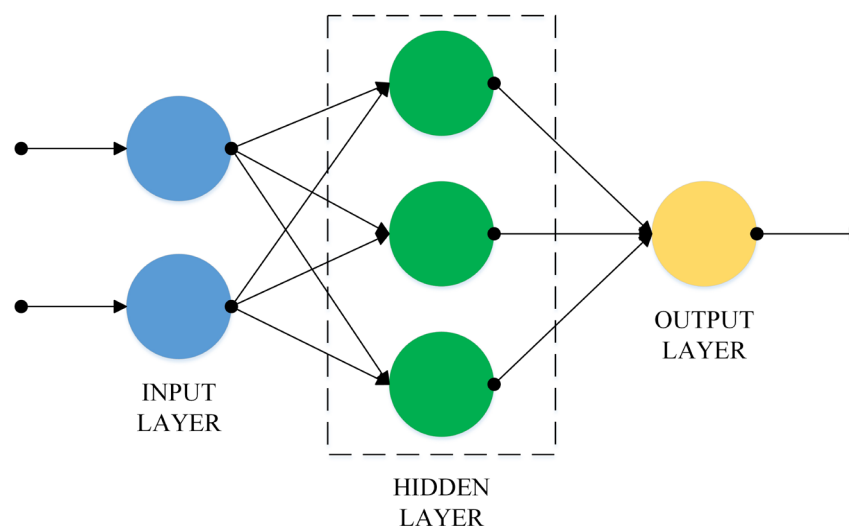


Figure 10. Artificial Neural Network architecture.

5.1.3. Fibonacci Series Algorithm

The Fibonacci Series (FS) MPPT algorithm uses principles from the Fibonacci sequence, a mathematical series where each number is the sum of the two preceding ones, to optimize the tracking of the MPP in PV systems. This algorithm systematically narrows down the search space for the MPP by using Fibonacci-based steps to adjust the operating point in a manner similar to interval reduction techniques [63,64]. Advantages of this approach include its simplicity and ease of implementation, as well as its efficiency in converging to the MPP by reducing the search space with each iteration. Additionally, it generally avoids oscillations around the MPP, leading to stable performance. However, disadvantages include its potential for slower convergence compared to more advanced algorithms and its sensitivity to changing conditions. The FS MPPT algorithm may require careful tuning of parameters to ensure optimal performance in varying environmental conditions [65]. Equation (5) provides the iterative sequence, with R representing the points on the PV curve used to track the MPP. Table 3 presents a comparison of the smart MPPT algorithms.

$$R_{(n+2)} = R_{(n+1)} + R_n, (n = 1, 2, 3, \dots \text{and } R_1 = R_2 = 1) \quad (5)$$

Table 3. Comparison of smart MPPT algorithms [29,53,66].

Tracking Algorithm	Sensed Parameters	Circuitry	Tracking Speed	Implementation Complexity	True MPPT	Accuracy	Cost
FLC	V, I	A/D	Slow	Low	Yes	Medium	Low
ANN	V, I	D	Medium	Medium	Yes	Medium	Medium
FS	G, T or I, T	D	Medium	Low	Varies	High	Low

5.2. Metaheuristic MPPT Techniques

Metaheuristic MPPT techniques provide advanced solutions for optimizing PV systems, especially in environments where traditional methods may fall short. Their ability to handle complexity, dynamic conditions, and multi-objective scenarios makes them a valuable tool in maximizing PV system performance [67]. They can be classified as optimization-based and bio-inspired MPPT algorithms. Figure 11 illustrates the general outline of the flowchart for metaheuristic MPPT techniques.

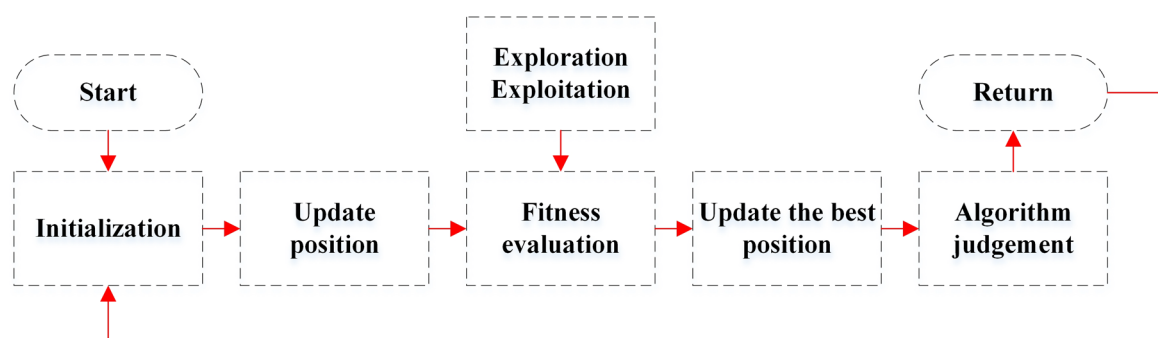


Figure 11. General outline of the flowchart for metaheuristic MPPT techniques.

5.2.1. Optimization-Based Algorithms

Optimization-based MPPT algorithms focus on solving the MPPT problem through mathematical and numerical optimization techniques. These algorithms often involve formulating the MPPT as an optimization task and using algorithms designed to find the optimal solution.

Particle Swarm Optimization Algorithm

The Particle Swarm Optimization (PSO) MPPT algorithm is an advanced technique used to maximize the power output of PV systems by mimicking the social behavior observed in flocks of birds or schools of fish [68]. In the flowchart of the PSO MPPT, as shown in Figure 12, a swarm of candidate solutions, or particles, converges on the MPP by iteratively adjusting the DC-DC converter's duty cycle based on the collective behaviors, where i refers to the iteration counter. Each particle adjusts its position (P_{best}) based on its own experience and that of its neighbors, gradually converging toward the optimal solution (G_{best}). Once the algorithm terminates, output the optimal duty cycle and the corresponding MPP. The main advantage of PSO is its ability to effectively handle complex, non-linear, and multi-modal optimization problems, often yielding superior performance. It is particularly useful in systems with dynamic or unpredictable environmental conditions. However, PSO can be computationally intensive, necessitating careful parameter tuning for optimal performance, and it often involves a more complex implementation than simpler traditional MPPT methods. The initial conditions and the specific design of the PSO parameters can affect the algorithm's effectiveness [69–71]. The following equations are used to update each particle's position and velocity. Here, x_i and Φ_i represent the position and velocity of each particle, respectively. k denotes the iteration count, w is the inertia weight, c_1 and c_2 are acceleration coefficients, and r_1 and r_2 are random values uniformly distributed between 0 and 1.

$$x_i^{k+1} = x_i^k + \Phi_i^{k+1} \quad (6)$$

$$\Phi_i^{k+1} = w\Phi_i^k + c_1r_1\{P_{best} - x_i^k\} - c_2r_2\{G_{best} - x_i^k\} \quad (7)$$

Grey Wolf Optimization Algorithm

The Grey Wolf Optimization (GWO) MPPT algorithm is based on the hunting behavior and social hierarchy of grey wolves [72]. Figure 13 depicts the flowchart of the GWO MPPT, where i represents the index of the individual wolf within the population and k represents the current iteration number. In this algorithm, a population of candidate solutions, represented as wolves, is used to search for the MPP of a PV system. The wolves are categorized into alpha, beta, delta, and omega roles, with alpha wolves leading the search and beta and delta wolves assisting them. The algorithm mimics the pursuit and encirclement strategies of wolves during hunting to explore and exploit the solution space effectively [73]. One of the main advantages of the GWO algorithm is its robustness and ability to converge quickly to the MPP, even in complex and dynamic conditions. It generally performs well in finding optimal solutions with fewer iterations compared to some other optimization algorithms. However, the GWO algorithm's computational complexity can be higher than simpler MPPT methods, making it less suitable for systems with limited processing power. The PV module output power is calculated using the voltage and current measurements for the grey wolves, which represent the duty ratios [74,75]. GWO behavior can be modeled by the following equations, where $\vec{X}$ and $\vec{X}_p$ represent the position vectors of the grey wolf and the prey, respectively. t denotes the number of iterations, while $\vec{A}$, $\vec{C}$, and $\vec{D}$ are the coefficient vectors. $\vec{a}$ is a parameter that decreases linearly from 2 to 0, and $\vec{r}_1$ and $\vec{r}_2$ are random vectors within the range of [0, 1].

$$\vec{D} = \vec{C} * \vec{X}_p(t) - \vec{X}_p(t) \quad (8)$$

$$\vec{X}(t+1) = \left| \vec{X}_p(t) - \vec{A} * \vec{D} \right| \quad (9)$$

$$\vec{A} = 2 * \vec{a} * r_1 - \vec{a} \quad (10)$$

$$\vec{C} = 2 * r_2 \quad (11)$$

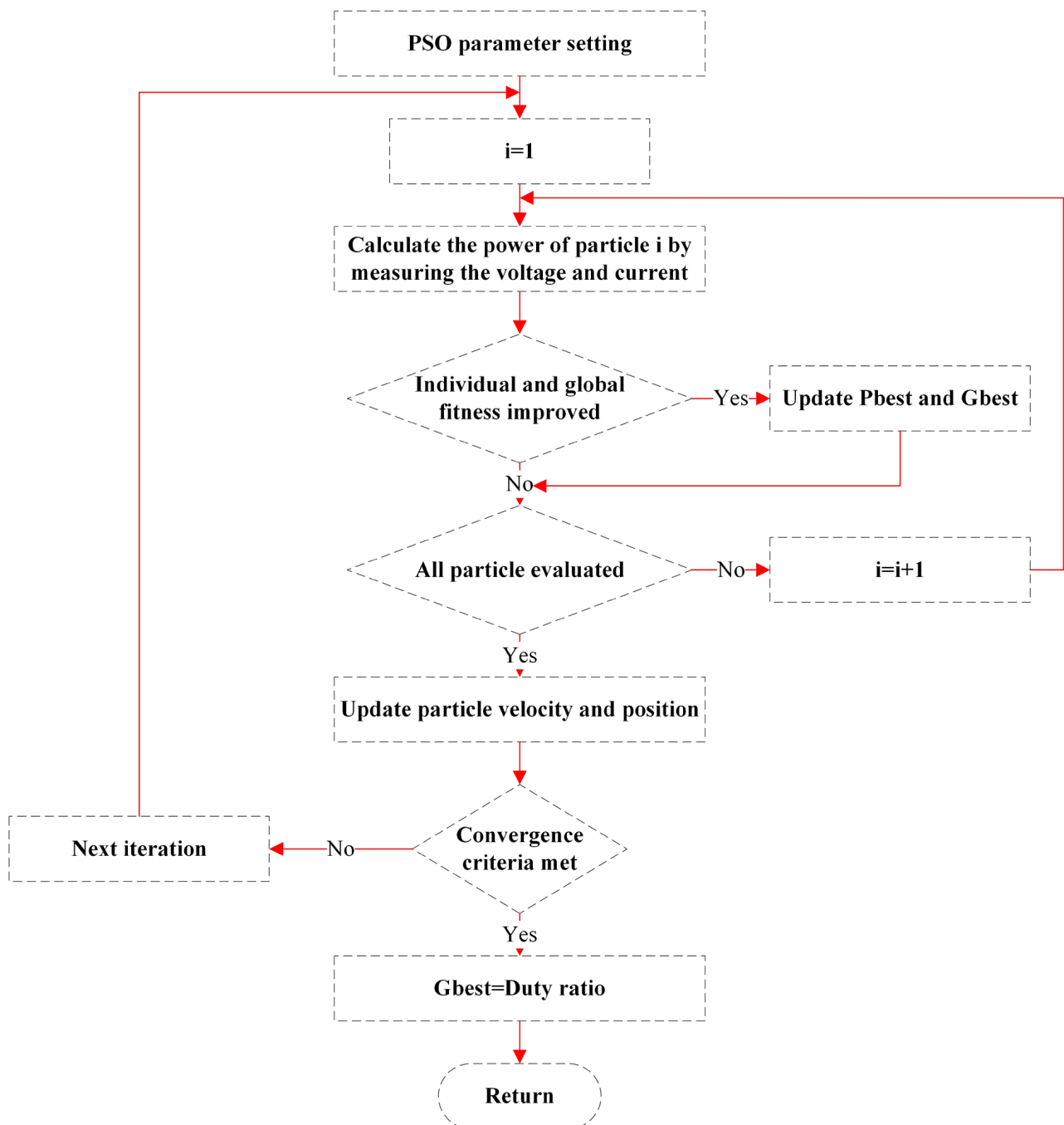


Figure 12. Flowchart of the PSO MPPT.

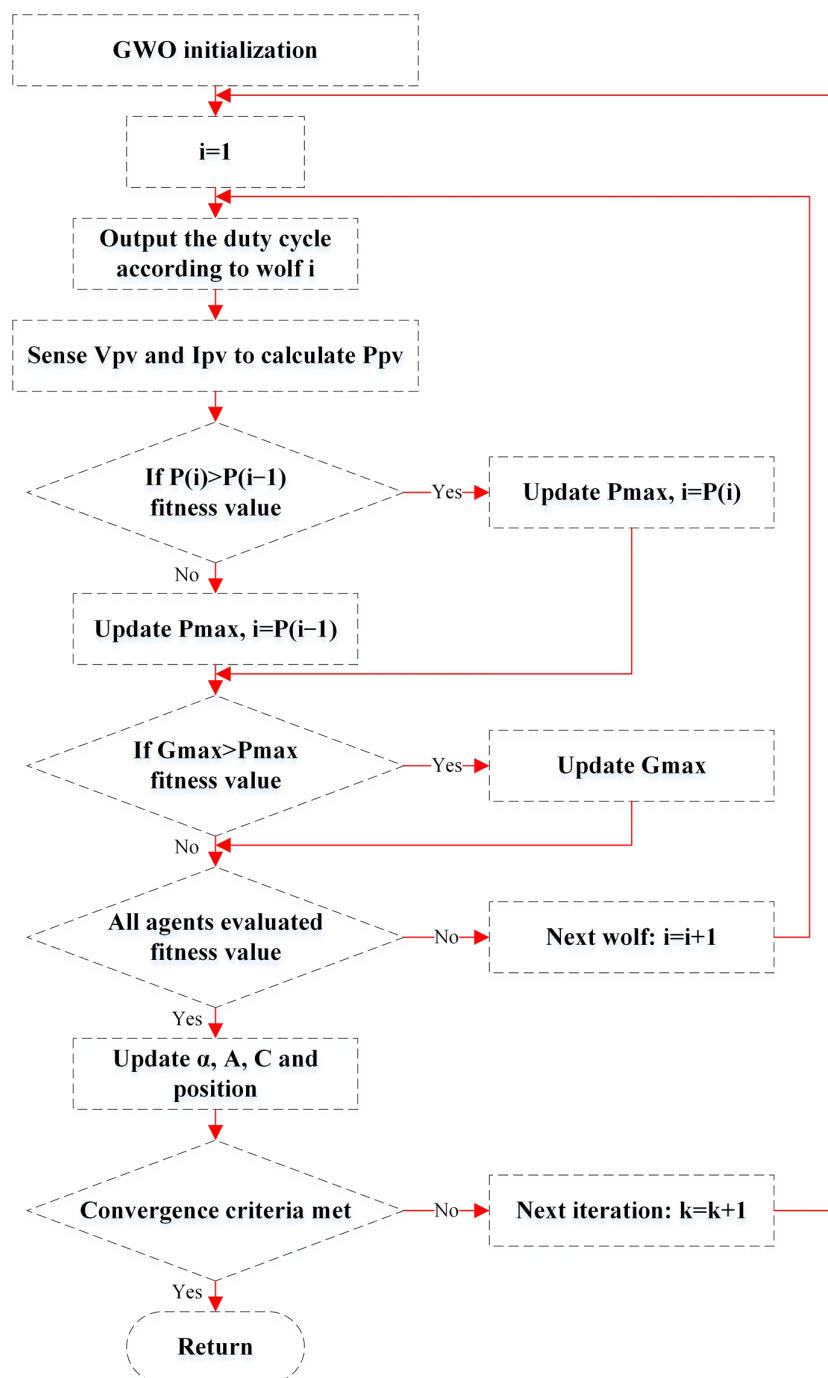


Figure 13. Flowchart of the GWO MPPT.

Ant Colony Optimization Algorithm

The Ant Colony Optimization (ACO) MPPT algorithm is inspired by the foraging behavior of ants, where ants deposit pheromones to identify successful paths and guide others [76]. Figure 14 displays the flowchart of the ACO MPPT, where M is the number of ants in the colony and K is the number of iterations. In this algorithm, the search for the MPP is modeled as a network of paths, with virtual “ants” exploring different solution routes and updating their pheromone trails based on the power output they encounter. Each ant explores different paths (duty cycles) and calculates the PV module output power. Successful paths that yield higher power result in increased pheromone levels on those paths. This approach allows the algorithm to effectively balance the exploration of new regions and the exploitation of known successful areas. Advantages of the ACO

algorithm include its robustness in handling dynamic conditions, ability to avoid local optima, and rapid convergence to the MPP by using collective intelligence. However, disadvantages include the potential for higher computational complexity and the need for careful parameter tuning to balance exploration and exploitation effectively [77,78]. In the equations given below, $G_i(x)$ represents the Gaussian kernel for the i -th dimension of the solution, while w represents the weight associated with the l -th term in the sum, $g_l^i(x)$ denotes the l -th sub-Gaussian function for the i -th dimension. l is the index that runs over the sum from 1 to K and x represents a specific input. μ_l^i and σ_l^i are the mean value and standard deviation for the i -th dimension of the l -th solution, respectively.

$$G_i(x) = \sum_{l=1}^K w_l g_l^i(x) \quad (12)$$

$$\pi_{li}(x) = \frac{1}{\sigma_l^i \sqrt{2\pi}} \exp\left(-\frac{(x - \mu_l^i)^2}{2\sigma_l^{i2}}\right) \quad (13)$$

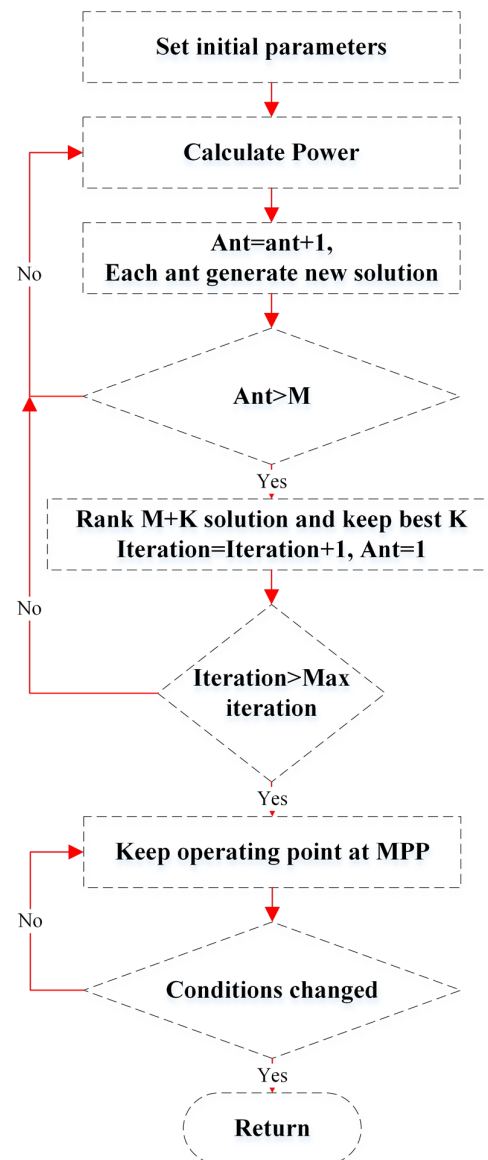


Figure 14. Flowchart of the ACO MPPT.

Artificial Bee Colony Algorithm

The Artificial Bee Colony (ABC) MPPT algorithm is inspired by the foraging behavior of honeybees to find optimal food sources [79]. Figure 15 shows the flowchart of the ABC MPPT. It begins by initializing a population of bees, each representing a potential duty cycle. Employed bees evaluate their current positions by measuring the PV module's output power and exploring neighboring positions for improvements. This iterative process continues until convergence to the MPP is reached [80]. The advantages of the ABC algorithm include its robustness in diverse and dynamic conditions, and its capability to avoid local MPP through a balance of exploration and exploitation. However, disadvantages may include computational overhead and the need for careful parameter tuning to ensure optimal performance [81]. The optimization-based MPPT algorithms are compared in Table 4.

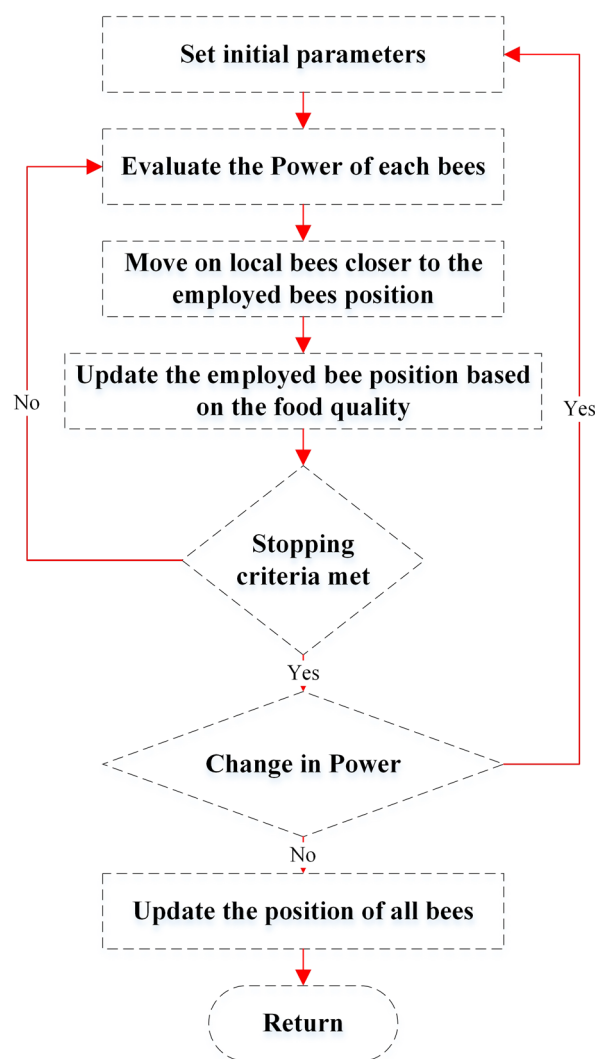


Figure 15. Flowchart of the ABC MPPT.

Table 4. Comparison of optimization-based MPPT algorithms [53,82].

Tracking Algorithm	Sensed Parameters	Circuitry	Tracking Speed	Implementation Complexity	True MPPT	Accuracy	Cost
PSO	V, I	D	Fast	Medium	Yes	High	Medium
GWO	V, I	D	Very Fast	Medium	Yes	High	Medium
ACO	V, I	D	Fast	Complex	Yes	Medium	High
ABC	V, I	D	Fast	Complex	Yes	Medium	High

5.2.2. Bio-Inspired Algorithms

Bio-inspired MPPT algorithms draw inspiration from natural processes and behaviors observed in animals and biological systems. These algorithms mimic these natural phenomena to solve the MPPT problem, often through swarm intelligence, evolutionary strategies, or other bio-inspired mechanisms [83].

Cuckoo Search Optimization Algorithm

The Cuckoo Search Optimization (CSO) MPPT algorithm is an advanced optimization technique inspired by the brood parasitism of cuckoo birds, where solutions are treated as cuckoo eggs laid in host nests [84]. Figure 16 indicates the flowchart of the CSO MPPT, where a is the scaling factor and n is the population size. This algorithm iteratively explores and exploits potential solutions to identify the optimal MPP by mimicking the natural behavior of cuckoos. Its advantages include robust performance in dynamic environments and a strong capability to avoid local optima, leading to faster convergence compared to traditional methods. However, it can be more complex and computationally intensive, requiring careful parameter tuning to achieve optimal results [85,86]. In the CSO algorithm, Lévy flights represent a random walk of cuckoo birds, defined by the following equations. Here, y is the probability density, λ represents the variance, and l denotes the flight length. x_i^t is the sample, while α indicates the step size. The variables t and i refer to the number of iterations and samples, respectively. The value of α is the step size, where α_0 is the initial step size. Additionally, $x_j^{(t)}$ and $x_i^{(t)}$ are the two samples. By using Lévy flights for step size, CSO achieves faster convergence.

$$y = l^{-\lambda}, \quad (1 < \lambda < 3) \quad (14)$$

$$x_i^{(t+1)} = x_i^t + \alpha \oplus \text{Lévy}(\lambda) \quad (15)$$

$$a = a_0(x_i^{(t)} - x_i^{(t)}) \quad (16)$$

Firefly Optimization Algorithm

The Firefly Optimization (FO) MPPT algorithm is inspired by the natural behavior of fireflies, which use light flashes to attract mates and communicate [87]. Figure 17 depicts the flowchart of the FO MPPT. In this algorithm, the search for the optimal MPP is modeled as a swarm of fireflies, where each firefly represents a potential solution and moves through the solution space based on the attractiveness of other fireflies, which is related to their brightness or power output. The advantages of this method include its ability to effectively explore a wide solution space, maintain a balance between exploration and exploitation, and adapt to varying conditions, leading to improved convergence and accuracy in reaching the MPP. However, the algorithm can be computationally demanding and may require fine-tuning of its parameters to achieve optimal performance, which can be a drawback compared to simpler MPPT techniques [88,89]. Following equations are used for the FO algorithm, where D represents the dimension, $x_{i,k}$ is the k -th dimension of the i -th firefly, $x_{j,k}$ is the k -th dimension of the j -th firefly, r_{ij} is the distance between the i -th and j -th fireflies, β_0 is the attractiveness at $r = 0$, e is the mathematical constant, γ is the absorption coefficient, and α is a random movement factor within the range of $[0, 1]$.

$$r_{ij} = \|x_i - x_j\| = \sqrt{\sum_{k=1}^D (x_{i,k} - x_{j,k})^2} \quad (17)$$

$$\beta(r_{ij}) = \beta_0 e^{-\gamma r_{ij}^2} \quad (18)$$

$$x_i = x_i + \beta_0 e^{-\gamma r_{ij}^2} (x_i - x_j) + \alpha(\text{rand} - 0.5) \quad (19)$$

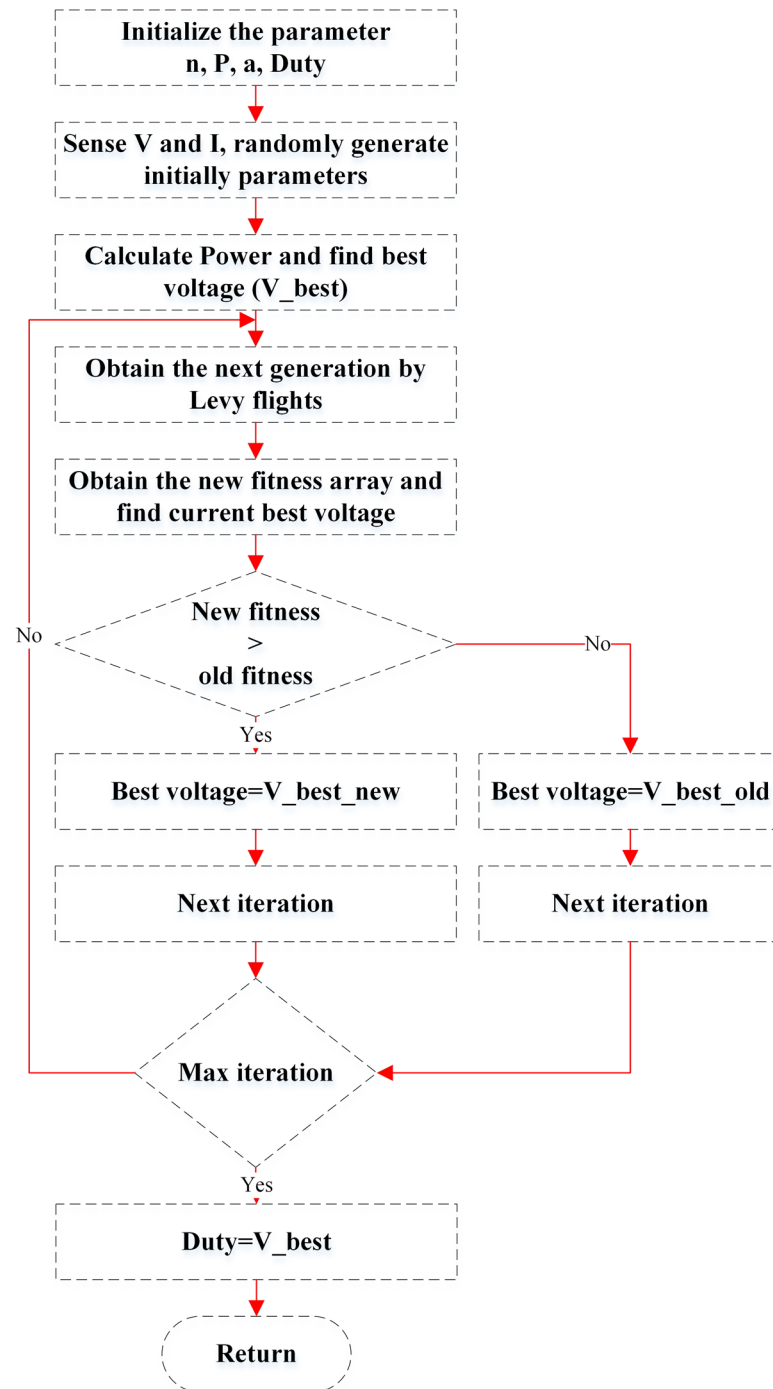


Figure 16. Flowchart of the CSO MPPT.

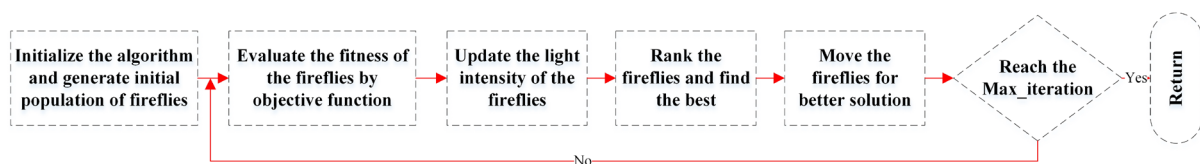


Figure 17. Flowchart of the FO MPPT.

Flower Pollination Algorithm

The Flower Pollination (FP) MPPT algorithm is inspired by the process of flower pollination and is used in MPPT to optimize power extraction from PV systems. Figure 18

displays the flowchart of the FP MPPT, where P refers to the probability factor and $Rand$ refers to a random number generated within a specified range. It operates by mimicking the biological processes of pollination and reproduction, exploring and exploiting the search space to find the MPP [90]. Advantages of the FP algorithm include its ability to handle complex, multi-dimensional optimization problems with fewer parameters to tune compared to some other algorithms. It is also robust against local optima due to its stochastic nature. Disadvantages include potentially higher computational complexity and convergence time compared to simpler algorithms, and it may require more careful tuning for specific applications to achieve optimal performance [91–93]. The following equations are used for the FP, where x_i^t refers to the pollen, g_{best} is the current best solution, γ is the scaling factor that controls the step size, and $L(\lambda)$ represents the step size based on Lévy flights. $\Gamma(\lambda)$ is the gamma function, applicable for large steps. x_i^t and x_j^t are the pollens from different flowers of the same plant species, and ε denotes the local search distributed uniformly within the range $[0, 1]$.

$$x_i^{t+1} = x_i^t + \gamma L(\lambda)(g_{best} - x_i^t) \quad (20)$$

$$L(\lambda) = \frac{\lambda \Gamma(\lambda) \sin(\frac{\pi}{2})}{\pi} - \frac{1}{s^{1+\lambda}}, s \gg s_0 > 0 \quad (21)$$

$$x_i^{t+1} = x_i^t + \varepsilon(x_j^t - x_k^t) \quad (22)$$

Genetic Algorithm

Genetic Algorithm (GA)-based MPPT algorithms are inspired by the principles of natural selection and evolution to optimize the power output of PV systems [94]. Figure 19 indicates the flowchart of the GA MPPT, where i is the generation number in the optimization process. The algorithm works by encoding potential solutions as chromosomes and applying genetic operations such as selection, crossover, and mutation to evolve over successive generations. The algorithm evaluates the performance of each solution based on a fitness function, which in this case is typically the power output of the PV system, and iteratively refines the solutions to approach the MPP. One of the main advantages of GA-based MPPT algorithms is their robustness and flexibility in exploring a wide range of possible solutions, making them effective for complex problems and environments with partial shading or rapid irradiance changes. However, GA can be computationally intensive and may require a considerable amount of time to converge to an optimal solution. Additionally, the performance of GA can be sensitive to the choice of parameters, such as mutation rates and population size, which may require careful tuning to achieve the best results [95,96]. Table 5 provides a comparison of bio-inspired MPPT algorithms.

Table 5. Comparison of bio-inspired MPPT algorithms [53,97].

Tracking Algorithm	Sensed Parameters	Circuitry	Tracking Speed	Implementation Complexity	True MPPT	Accuracy	Cost
CSO	V, I	D	Fast	Low	Yes	High	High
FO	V, I	D	Very Fast	Medium	Yes	High	Medium
FP	V, I	D	Very Fast	Medium	Yes	High	Medium
GA	V, I	D	Fast	Complex	Yes	Medium	Medium

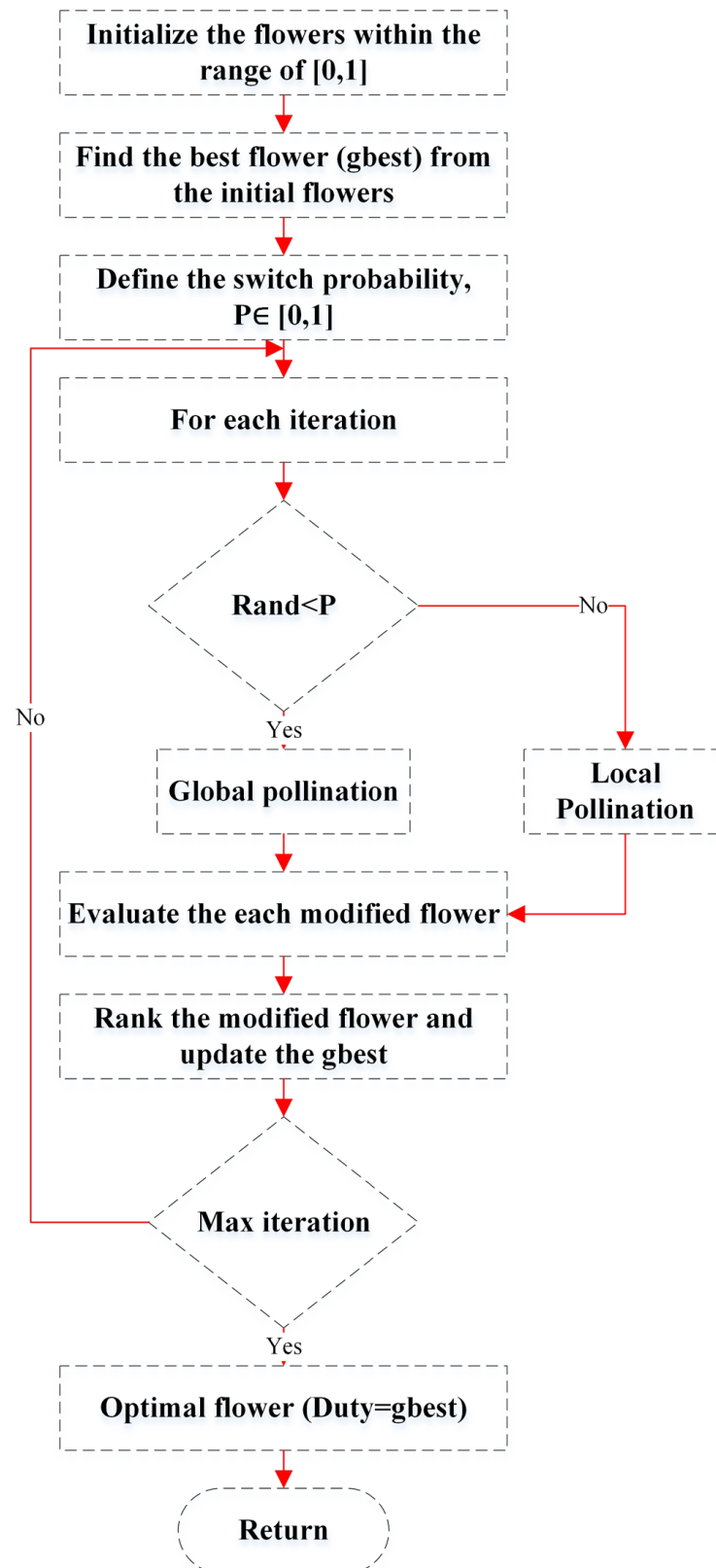


Figure 18. Flowchart of the FP MPPT.

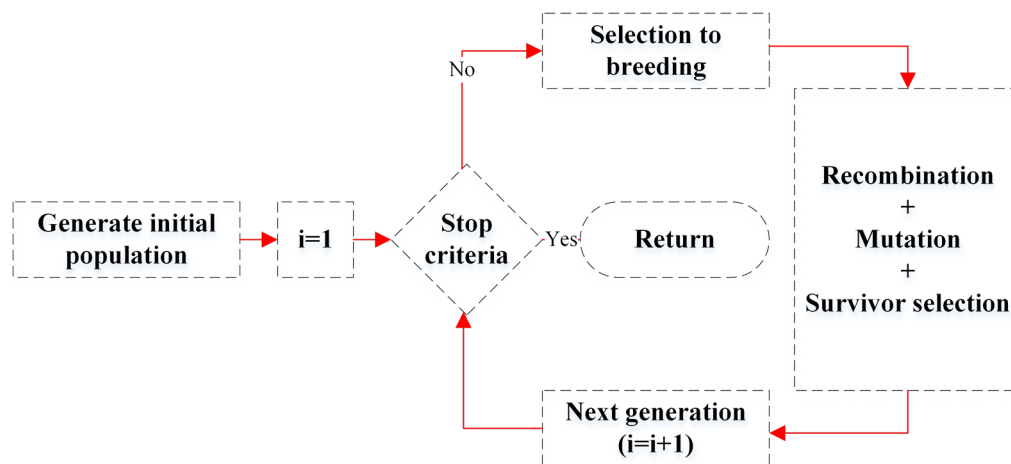


Figure 19. Flowchart of the GA MPPT.

5.3. Hybrid MPPT Techniques

Hybrid MPPT techniques combine multiple MPPT algorithms to use their individual strengths and mitigate their weaknesses. They integrate different MPPT strategies to create a more robust tracking system to enhance the performance and adaptability of PV systems, particularly in environments with complex or rapidly changing conditions [29,66].

5.3.1. PSO-P&O Algorithm

The PSO-P&O hybrid MPPT algorithm is an advanced method that combines the global search capabilities of PSO with the local tracking accuracy of P&O [98]. In this hybrid approach, PSO is used to explore a broad range of potential MPPs by adjusting the operating voltage of the PV system and optimizing the search process based on the particles' performance. Once the PSO algorithm converges to a near-optimal region, the P&O method is employed to finely adjust the operating point and precisely track the MPP with high accuracy. Advantages of the PSO-P&O hybrid algorithm include its improved ability to handle complex and dynamic environmental conditions due to PSO's robust global search capability, which helps in avoiding local maxima and ensuring better convergence to the actual MPP. The P&O component enhances the precision of power tracking once a promising region is identified, providing efficient operation under varying conditions. Disadvantages include increased computational complexity and implementation challenges. The PSO component requires significant computational resources to perform the global search, which can be demanding in terms of processing power and time. Additionally, managing the transition between PSO and P&O, as well as tuning their parameters effectively, can be complex and may require careful calibration to balance the global search and local tracking benefits [34,36].

5.3.2. GWO-P&O Algorithm

The GWO-P&O hybrid MPPT algorithm integrates the global optimization capabilities of GWO with the local search accuracy of P&O. In this hybrid approach, the GWO algorithm is initially employed to explore a broad range of potential MPPs by mimicking the hunting behavior of grey wolves. GWO effectively identifies a promising region for the MPP due to its strong global search ability. Once this region is determined, the P&O method takes over to fine-tune the operating point with high precision, adjusting the voltage to ensure that the system operates exactly at the MPP [34,69]. Advantages of the GWO-P&O hybrid algorithm include enhanced performance in complex environmental conditions, as GWO's global search capabilities help avoid local maxima and improve convergence to a more accurate

MPP. The P&O component then refines the power tracking, offering precise and efficient operation. Disadvantages involve increased computational requirements and complexity. GWO can be computationally intensive due to the need for multiple iterations to perform global optimization, which might demand more processing power and time [36].

5.3.3. FLC-P&O Algorithm

The FLC-P&O hybrid MPPT algorithm combines FLC with the P&O method to optimize MPPT in PV systems. The FLC dynamically adjusts the perturbation step size based on system inputs such as changes in power and voltage, improving the system's responsiveness to varying environmental conditions like fluctuating irradiance and temperature. Once the fuzzy controller sets the optimal step size, the P&O method fine-tunes the voltage to track the MPP by perturbing and observing power changes. This hybrid approach enhances tracking efficiency, reduces oscillations around the MPP, and adapts better to variable conditions. However, it introduces increased complexity, requires careful tuning of fuzzy rules, and demands more computational power, making it more resource-intensive compared to simpler algorithms [99].

5.3.4. GA-P&O Algorithm

The GA-P&O hybrid MPPT algorithm combines the global search capabilities of GA with the local optimization precision of P&O to enhance the tracking of the MPP in PV systems. GA initially explores a broad range of potential MPPs by mimicking natural selection, identifying a promising MPP region. Once this region is determined, the P&O algorithm fine-tunes the operating point for precise power tracking. This hybrid approach offers the advantage of avoiding local maxima and providing accurate MPP tracking, especially in complex or rapidly changing environmental conditions. However, it comes with the drawback of increased computational complexity, longer convergence times, and higher implementation costs due to the iterative nature of GA and the need for more sophisticated hardware and tuning. Despite these challenges, the GA-P&O hybrid is beneficial for systems requiring robust performance in dynamic conditions [100].

5.3.5. Adaptive Neuro-Fuzzy Inference System-Based Algorithm

The Adaptive Neuro-Fuzzy Inference System (ANFIS)-based MPPT algorithm uses the combined strengths of neural networks and fuzzy logic to optimize the power output of PV systems [101]. ANFIS uses a hybrid approach, incorporating the learning capabilities of neural networks with the rule-based reasoning of fuzzy logic to model and predict the MPP. The algorithm adapts to varying environmental conditions by training a fuzzy inference system with historical and real-time data to improve its ability to forecast the optimal operating point of the PV system. Advantages of the ANFIS-based MPPT algorithm include its adaptability to complex and dynamic conditions. By learning from data, ANFIS can effectively handle non-linear and unpredictable variations in solar irradiance and temperature, making it highly effective in diverse environments. Additionally, the fuzzy logic component allows for a more intuitive handling of uncertain or imprecise inputs, enhancing the system's robustness. Disadvantages include the computational complexity and the need for extensive training data. ANFIS requires significant computational resources to train the fuzzy inference system and adjust its parameters, which can be demanding in terms of processing power and time. Moreover, the quality of the MPPT performance is heavily dependent on the quality and quantity of the training data; inadequate or unrepresentative data can lead to suboptimal performance [102]. Table 6 compares various hybrid MPPT algorithms.

Table 6. Comparison of hybrid MPPT algorithms [36].

Tracking Algorithm	Sensed Parameters	Circuitry	Tracking Speed	Implementation Complexity	True MPPT	Accuracy	Cost
PSO-P&O	V, I	D	Fast	Complex	Yes	High	Medium
GWO-P&O	V	D	Medium	Complex	Yes	High	Medium
FLC-P&O	V, I	D	Fast	Medium	Yes	Medium	Medium
GA-P&O	V, I	D	Fast	Complex	Yes	High	High
ANFIS	V, I	D	Fast	Complex	Yes	Medium	High

6. Discussion

Both traditional and advanced MPPT algorithms play crucial roles in optimizing PV systems, each offering distinct benefits and addressing different needs within the field of renewable energy. Due to their simplicity and ease of use, traditional MPPT algorithms like P&O and IC have long been the basis of solar power optimization. They are particularly effective in stable or predictable environments, where their basic approaches can efficiently track the MPP with minimal computational resources. However, they may struggle with rapidly changing environmental conditions or in systems requiring high precision. In contrast, advanced MPPT algorithms are designed to handle the complexities of modern, dynamic environments. These algorithms offer enhanced performance through their ability to adapt to varying conditions, optimize across multiple objectives, and improve tracking accuracy. While they come with increased computational demands and complexity, their advanced capabilities make them well-suited for large-scale PV systems, smart grids, and other sophisticated applications. As a result, in environments where traditional methods are inadequate, smart, optimization and bio-inspired-based tracking methods have surfaced. Moreover, there is a tendency to apply such combinations of critical and operational advances as PSO-P&O and GWO-P&O, which seem to be effective due to the combination of the global applicability of modern methods and the local effectiveness of traditional methods.

MPPT algorithms typically use electrical or atmospheric data to find the MPP. Analog MPPT algorithms rely on op-amps or simple circuits, which are inexpensive, simple, and require no programming, controllers, or memory. However, they lack flexibility. Digital-based algorithms, which include a controller or processor, can be modified through changes in code or flowcharts, but this increases complexity.

There is a trade-off between MPPT algorithm complexity and tracking efficiency. Traditional methods like P&O and IC are simple with low computational needs but may be less efficient under rapidly changing or shaded conditions. Advanced techniques provide better tracking in non-ideal conditions but require more computational power and hardware. The choice depends on the PV system's application and constraints. For systems with limited resources or cost concerns, simpler methods are preferable. However, for critical applications where efficiency is key and resources are available, advanced techniques offer significant benefits despite their complexity and cost.

Partial shading can significantly impact the performance of PV systems, as it creates multiple LMPPs on the power–voltage curve, confusing traditional MPPT algorithms like P&O and IC. These traditional methods often fail to identify the true global MPP under partial shading, leading to inefficient energy harvest. Advanced MPPT algorithms, such as metaheuristic techniques, are better equipped to distinguish between local MPPs and global MPP, ensuring more accurate tracking and maximizing energy output. These algorithms offer superior performance, especially in complex shading conditions. Table 7 classifies the performance of MPPT algorithms with respect to the computational requirements and overall efficiency under uniform and PSC into low, lower-medium, medium, upper-medium, and high levels.

Table 7. Performance of MPPT algorithms [25,27,29,34,37,38].

Type	Tracking Algorithm	Computational Requirements	Efficiency in Uniform Shading	Efficiency in Partial Shading
Traditional	P&O	Low	Medium	Low
	IC	Lower-Medium	Medium	Low
	CV	Low	Lower-Medium	Low
	LT	Lower-Medium	Medium	Medium
	HC	Low	Medium	Low
Advanced	FLC	High	High	Medium
	ANN	Medium	High	High
	FS	High	High	Medium
	PSO	Medium	High	High
	GWO	Medium	High	High
	ACO	High	High	High
	ABC	Medium	Upper-Medium	High
	CSO	Medium	High	High
	FO	Medium	High	High
	FP	Medium	High	High
	GA	High	High	High
	PSO-P&O	High	High	Medium
	GWO-P&O	High	High	Medium
	FLC-P&O	High	Upper-Medium	Medium
	GA-P&O	High	Upper-Medium	Medium
	ANFIS	High	Upper-Medium	High

In the context of hotspot phenomena, traditional and advanced MPPT algorithms behave differently. Traditional algorithms are less effective at detecting and mitigating hotspots, often failing to identify localized temperature variations that can cause reduced efficiency or damage to the PV module. In contrast, advanced MPPT algorithms adapt better to environmental changes, optimizing power output more effectively and preventing overheating in specific regions of the PV module, thus reducing the risk of hotspots.

Many MPPT algorithms studied are successfully used in commercial products, such as MPPT-controlled battery charge controllers, string inverters, and solar tracking systems. However, these algorithms are primarily found in academic literature, with most studies conducted in simulation environments. Complex MPPT algorithms are rarely integrated into real-life applications through rapid prototyping and application.

Artificial Intelligence (AI) and Machine Learning (ML)-based MPPT algorithms require data sets to function effectively, gaining predictive capabilities through training. However, not all data sets may match the parameters of every solar energy system. For safe operation, the training data must cover the specific parameters of the system in which the algorithms will be applied, ensuring that the minimum, maximum, and optimum values of electrical and environmental data align with the training data and test environment. Additionally, since ML and AI-based algorithms demand intensive processing, they require a controller (regulator or processor) and memory components.

The emergence of hybrid algorithms is due to the inability of existing algorithms to provide the desired performance. An algorithm efficient in one area may be less effective in another. For example, traditional P&O and IC MPPT algorithms quickly reach the global MPP point but suffer from oscillations due to their fixed-step nature. Modified algorithms reduce oscillations but take longer to reach the global MPP. Hybrid algorithms combine the strengths of multiple MPPT algorithms, achieving both fast tracking and minimal oscillation by ensuring their harmonious operation. Table 8 presents a comparison of traditional and advanced MPPT algorithms, highlighting their advantages and disadvantages.

Table 8. Comparison of traditional and advanced MPPT algorithms [53,103,104].

Type	Tracking Algorithm	Advantages	Disadvantages
Traditional	P&O	Simple to implement.	Can oscillate around the MPP.
	IC	Better in varying conditions.	More complex and computational.
	CV	Straightforward for stable systems.	Ineffective with changing conditions.
	LT	Easy with known conditions.	Not adaptive to changes.
	HC	Simplicity.	Can oscillate around the MPP.
Advanced	FLC	Handles uncertainties well.	Complex rule setup required.
	ANN	Adapts to complex behaviors.	Needs extensive training data.
	FS	Efficient for specific problems.	May not always converge well.
	PSO	Effective exploration.	Computationally expensive.
	GWO	Handles complex conditions.	Requires parameter tuning.
	ACO	Good for dynamic systems.	Slow convergence.
	ABC	Exploration and exploitation.	Sensitive to parameter settings.
	CSO	Efficient for complex spaces.	High computational cost.
	FO	Effective for multi-modal issues.	Expensive and requires tuning.
	FP	Mimics natural processes.	Less effective in discrete problems.
	GA	Flexible and robust.	Slow convergence.
	PSO-P&O	Exploration with simplicity.	Inherits P&O's oscillation issues.
	GWO-P&O	Merges global search with P&O.	Complex and computationally intense.
	FLC-P&O	Dynamic step-size adjustment.	Increased complexity, needs tuning.
	GA-P&O	Precise tracking.	High computational cost.
	ANFIS	Robust and adaptive.	Requires significant training.

7. Current Trends and Future Directions

Current trends in MPPT algorithms are increasingly focused on integrating AI and ML to enhance adaptability and performance under variable conditions. Hybrid approaches, which integrate both traditional and advanced techniques, are gaining popularity due to their ability to combine global search capabilities with local tracking accuracy. Real-time adaptive MPPT systems are on the rise, allowing for dynamic adjustments [105]. Looking ahead, future developments are expected to include the application of advanced AI-based techniques, the potential use of quantum computing for complex optimizations, integration with smart grids, and improved environmental and economic optimization, all aimed at achieving more efficient and responsive PV systems. MPPT algorithms may evolve as a result of these advances [103,106]. Table 9 provides a summary of the current trends and applications for both traditional and advanced MPPT algorithms.

Deep learning and reinforcement learning are methods used in the optimization of solar PV systems. Large data sets increase the adaptation of deep learning. Thus, this makes MPPT algorithms for solar PV systems more sensitive. In a study conducted at Bilecik Seyh Edebali University [107], it was shown that artificial intelligence-based MPPT systems using deep learning exhibited more efficient performance than classical MPPT algorithms. Reinforcement learning, on the other hand, provides better adaptation to environmental changes. In a simulation study [108], it was seen that the reinforcement learning-based MPPT controller gave very good results under different operating conditions, and varying temperatures, radiation intensities, and electrical loads in the simulation.

Table 9. Current trends and applications for traditional and advanced MPPT algorithms [105,106].

Type	Current Trends	Applications
Traditional	Widely used; some hybrid versions are emerging.	Common in small to medium-scale PV systems.
	Increasing popularity in dynamic conditions.	Used in residential and commercial PV systems.
	Using with predefined tables for known conditions	Applied in scenarios with stable solar conditions.
	Less common; used in stable environments.	Limited to specific scenarios with stable conditions.
Advanced	Growing integration with smart grids and complex systems.	Employed in advanced PV systems and smart grids.
	Increasing use due to advancements in AI and data analytics.	Used in predictive maintenance and optimization.
	Rising interest in high-dimensional and complex optimization problems.	Applied in large-scale PV systems and energy management.
	Focus on balancing exploration and exploitation.	Used in logistics and dynamic energy systems.
	Effective solutions sought for continuous optimization and complex control.	Applied in various engineering and energy optimization tasks.

Advanced MPPT algorithms often require substantial computational resources due to their complex mathematical operations, iterative processes, and real-time data analysis. This can lead to increased processing time and power consumption, especially in embedded systems with limited processing power or energy constraints. Balancing accuracy and performance is crucial, as these algorithms require more sophisticated hardware for fast data processing and decision making. This increases system cost and complexity, making them less practical for small-scale or cost-sensitive applications. Developing algorithms with minimal hardware while maintaining performance is a challenge. Additionally, these algorithms must adapt quickly to changing environmental conditions, ensuring efficiency and stability in real-time systems without delays, which is difficult for computationally intensive methods.

There are indeed unresolved challenges that need further exploration. One is improving their robustness under dynamic and non-uniform conditions, such as rapidly changing solar radiation, partial shading, and temperature fluctuations. While advanced techniques perform well in stable conditions, they struggle in real-world scenarios. Another area is optimizing computational complexity. AI-based algorithms are efficient but can be limiting in low-cost or low-power systems. Future research could focus on simplifying these methods without sacrificing performance. Additionally, more work is needed on hybrid MPPT techniques that combine strengths from multiple approaches and improve real-time adaptability to environmental changes.

8. Conclusions

This paper discusses widely used important MPPT methodologies, including traditional and advanced techniques, in solar PV systems based on their performance under uniform and non-uniform irradiance conditions. While traditional MPPT algorithms may still be effective under uniform insolation, they often fail to achieve optimal performance in complex environments, such as partial shading. Advanced techniques, though more complex and costly, offer greater efficiency and accuracy in tracking the true MPP. The choice between traditional and advanced MPPT algorithms ultimately depends on the specific requirements of the PV system, including its scale, environmental variability, and the desired level of optimization. The comprehensive comparison of various MPPT methods provided in this paper offers valuable insights, enabling researchers and practitioners to make informed decisions when selecting the most suitable approach for specific PV applications.

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